Term	Definition	Key Attribute
System	The PathSG web application. It helps parents and students in Singapore plan education from preschool through university in one place.	— (singleton; not applicable)
Guest User	A person who has not logged in. A Guest User can browse the School Directory, view School profiles and read School Reviews. A Guest User cannot generate roadmaps, bookmark Schools or post Reviews.	None — no stored identifier
Authenticated User	A person who has registered and logged in with a username and password. An Authenticated User can save a Student Profile, bookmark Schools, generate and save a Roadmap, and post a School Review.	Username
Administrator	A staff user with the Admin role. An Administrator can open the Admin Panel to update or export the School Dataset.	Username (Role = Admin)
Access Token (JWT)	A short-lived JSON Web Token given to a User after login. It is sent in the Authorization header of each protected request and expires after 5 minutes.	Token string (JWT)
Refresh Token	A single-use token that gives a new Access Token and Refresh Token when the Access Token expires. It is valid for 7 days. The server stores only a hash of it, and the browser keeps it in session storage until the tab is closed.	Token hash
Rate Limit	A cap on repeated requests. A username may make at most 5 login attempts in 5 minutes from one client IP address, and a User account at most 10 AI roadmap requests in 5 minutes. Extra requests are rejected with HTTP 429.	Username + client IP address (login); User ID (AI roadmap)
Student Profile	The academic and location information an Authenticated User gives to get personal roadmap suggestions. It holds the Target Education Level, Academic Scores and Postal Code. It does not include personal details such as full name, birth date or NRIC/FIN number.	User ID Previous Academic Score Postal Code
Target Education Level	The next transition a student is planning for: No school to Preschool, Preschool to Primary, Primary to Secondary, Secondary to Post-Secondary, or Post-Secondary to University. It decides which Academic Score fields are shown and which stage a Roadmap is made for.	Level label
Academic Score	A previous academic result in the Student Profile. PSLE AL must be a whole number from 4 to 32. O-Level L1R5 must be a whole number from 2 to 54. O-Level ELR2B2 must be a whole number. Polytechnic GPA must be from 0.00 to 4.00. A-Level Rank Points must be from 0 to 90.	Score type
Postal Code	A 6-digit Singapore residential postal code. It is saved in the Student Profile and is the starting point for all distance calculations. Until a Postal Code is saved, the directory starts from a sample code (556000) so that distances can be shown; the Saved Institutions page shows no distance until one is saved.	6-digit Postal Code

Term	Definition	Key Attribute
Postal District	One of 28 areas of Singapore. The first two digits of a Postal Code select the district. The centre of the district is used as the starting point for distances. The System does not use an external mapping service to work out distance.	District number
School Governance Type	A label showing how a School is run, such as Government, Government-Aided or Independent.	Governance Type label
Educational Tier	A label for the level of a School: Primary, Secondary, Post-Secondary or University. A School that covers more than one level, such as an integrated school, belongs to every tier it covers. Preschool is used for roadmap stages only; the school data has no preschools yet, so a Preschool roadmap lists no school.	Tier label
School (Institution)	A school at any level, such as Primary, Secondary, Junior College, Polytechnic, ITE or University.	School Name / School ID
Course (Programme)	A specific diploma, degree or subject combination offered by a School, with its fees.	Course Name
School Fee	The cost of a Course, shown separately for Singapore Citizen, Permanent Resident and International Student.	Citizenship Status
Financial Support Scheme	Money help from the government or a School, such as the MOE Financial Assistance Scheme.	Scheme Name
Scholarship	A money award linked to a School or Course, with its own rules on who can apply.	Scholarship Name
Historical Entry Requirement	Past data used to judge how competitive a School is. Examples are the 2017 PSLE T-score cut-off, places taken up to P1 Phase 2B, JAE cut-off points and GPA range.	Requirement Year / Score Range
Roadmap	A plan of up to 3 Schools, in order of preference, for the Target Education Level. A Roadmap is made in two versions at the same time: a Rule-Based Plan and an AI-Generated Plan.	Stage / School IDs
Rule-Based Plan	A Roadmap ranked by fixed rules for the chosen stage. It is ready within 2 seconds. It is not available for Preschool or University stages.	Stage / Ranking Priority
AI-Generated Plan	A Roadmap made by an AI service (OpenAI) from the Student Profile and a list of valid Schools. It is made separately, so if it fails the Rule-Based Plan is not affected. It must finish within 15 seconds.	Stage
Ranking Priority	The choice of what the Rule-Based Plan ranks by first: Distance or Grade. Grade is the default. The choice is offered only for Primary to Secondary and Secondary to Post-Secondary. P1 Registration ranks by distance only.	Distance / Grade
Acceptable Distance Limit	The furthest distance a user accepts when ranking by Grade, from 1 km to Any distance. Schools beyond the limit are left out before ranking.	Distance in km
Saved Roadmap	A stored copy of a Roadmap. Each account can keep only one. Saving a new one replaces the old one after the user confirms. The user can edit its School choices or delete it.	User ID

Term	Definition	Key Attribute
Bookmarked School	A School that an Authenticated User has saved to a list of favourites. It is shown on the Saved Institutions page with its distance from the user's Postal Code.	User ID + School ID
School Review	A written comment of up to 200 words that an Authenticated User posts about a School. Each account can post one Review per School. The user can edit or delete their own Review.	Review ID
Location Map	A map shown on the School profile and in each Roadmap panel. It has a pin for each School and a house pin at the centre of the reference Postal District, the point distances are measured from. Map tiles are loaded from OneMap.	School ID
School Dataset	The set of raw CSV files from data.gov.sg that hold School information. The Administrator downloads them and uploads them in the Admin Panel. The System does not fetch them automatically.	School Key
Dataset Update	An action by an Administrator that cleans the uploaded CSV files and writes them to the database in a single transaction. If any error occurs, all changes are rolled back. Only one update can run at a time.	Update date / time
Dataset Export	An action by an Administrator that downloads all School records as one CSV file named pathsg_schools_YYYYMMDD.csv. It never includes user accounts, saved roadmaps or reviews.	File name